

COMPANION
to

ChatGPT Teaches TikZ

Chapter 10: Arrow Tips and Edge Labels (I)

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10.1 What This Companion Adds

Chapter 10 refined two parts of a connection: the arrow tip and the label. This Companion develops a small, consistent arrow vocabulary and a systematic way to place labels on straight and curved paths. The goal is precision without visual noise. A new arrowhead or label position should clarify mathematical meaning, not merely change appearance.

Design

Choose one ordinary arrow convention for a diagram. Introduce a second arrowhead only when it represents a genuinely different kind of relationship.

10.2 An Arrow-and-Label Vocabulary

Form or option	Purpose
<code>arrows.meta</code>	Loads the modern arrow library.
<code>--{Stealth}</code>	Places a Stealth tip at the end.
<code>{Stealth}--</code>	Places a Stealth tip at the start.
<code>{Stealth}{Stealth}</code>	Places tips at both ends.
<code>pos=.25</code>	Places a label one-quarter along the path.
<code>midway</code>	Places a label at the midpoint.
<code>near start, near end</code>	Places a label near an endpoint.
<code>sloped</code>	Rotates a label to follow the path.
<code>auto</code>	Lets TikZ choose a side of the path.
<code>swap</code>	Reverses the side chosen with <code>auto</code> .

Load the library in the preamble:

```
\usetikzlibrary{arrows.meta}
```

10.3 Modern Arrow Tips

The most useful general-purpose form is

```
\draw[-{Stealth}] (0,0) -- (4,0);
```

See



The braces belong to the modern arrow-tip syntax. They become especially important when the tip has options.

10.3.1 Tips at the Start and at Both Ends

```
\draw[Stealth-] (0,0) -- (4,0);  
\draw[Stealth-Stealth] (0,-1) -- (4,-1);
```

The direction of the path still runs from the first coordinate to the second. The arrow-tip specification decides which ends display tips.

10.3.2 Controlling Tip Size

Use a modest explicit size only when the ordinary tip is unsuitable.

```
\draw[-Stealth[length=4mm,width=3mm]]  
  (0,0) -- (4,0);
```

Avoid enlarging tips until they dominate the lines or labels.

10.4 Positioning a Label Along a Path

A path label is a node written inside the path command.

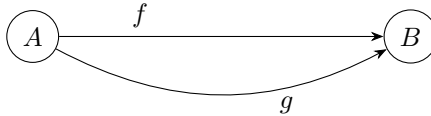
```
\draw[-Stealth] (A) -- node[midway,above] {f} (B);
```

Numerical positions give finer control:

```
\draw[-Stealth] (A) -- node[pos=.25,above] {f} (B);  
\draw[-Stealth] (A) -- node[pos=.75,below] {g} (B);
```

The number runs from 0 at the start to 1 at the end.

See



Use `near start` and `near end` when an exact fraction is not important.

10.5 Choosing the Side of a Path

On horizontal paths, `above` and `below` are direct and clear. On paths with changing directions, `auto` can choose a side relative to the path.

```
\draw[-{Stealth}] (A) -- node[auto] {$f$} (B);  
\draw[-{Stealth}] (A) -- node[auto,swap] {$g$} (B);
```

The second label appears on the opposite side from the first.

Common Mistake

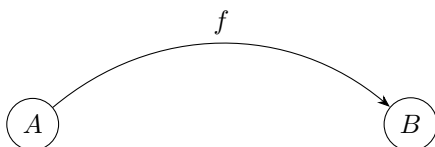
Do not place the label after the path has ended. The label node must occur between the starting point and the endpoint inside the same path command.

10.6 Labels on Curved Paths

The option `sloped` rotates the label so that its baseline follows the local direction of the curve.

```
\draw[-{Stealth}] (A)
  to[out=40,in=140]
  node[midway,sloped,above] {$f$}
  (B);
```

See



Use `sloped` only when rotation improves reading. A nearly vertical or strongly curved label may be clearer if it remains horizontal.

10.7 One Convention, Several Arrows

Collect a shared arrow convention in a style.

```
\tikzset{
  map/.style={-{Stealth},thick}
}
```

Then every ordinary mapping uses `map`. If the arrow tip changes, edit one definition rather than every path.

10.8 Worked Project: A Labeled Diagram

The project uses one arrow convention and places every label according to the direction of its path.

10.8.1 Stage 1: Place Four Objects

```
\node[draw,circle] (A) at (0,2) {$A$};  
\node[draw,circle] (B) at (4,2) {$B$};  
\node[draw,circle] (C) at (0,0) {$C$};  
\node[draw,circle] (D) at (4,0) {$D$};
```

10.8.2 Stage 2: Add the Square

```
\draw[map] (A) -- node[above] {$f$} (B);  
\draw[map] (A) -- node[left] {$g$} (C);  
\draw[map] (B) -- node[right] {$h$} (D);  
\draw[map] (C) -- node[below] {$k$} (D);
```

10.8.3 Stage 3: Add a Curved Comparison

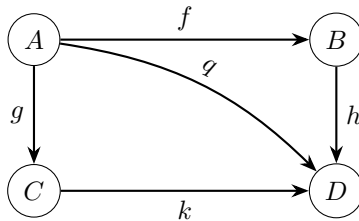
```
\draw[map] (A) to[bend left=18]  
node[midway,sloped,above] {$q$} (D);
```

10.8.4 The Complete Picture

```
\begin{tikzpicture}[
  map/.style={-{Stealth},thick}
]
  \node[draw,circle] (A) at (0,2) {$A$};
  \node[draw,circle] (B) at (4,2) {$B$};
  \node[draw,circle] (C) at (0,0) {$C$};
  \node[draw,circle] (D) at (4,0) {$D$};

  \draw[map] (A) -- node[above] {$f$} (B);
  \draw[map] (A) -- node[left] {$g$} (C);
  \draw[map] (B) -- node[right] {$h$} (D);
  \draw[map] (C) -- node[below] {$k$} (D);
  \draw[map] (A) to[bend left=18]
    node[midway,sloped,above] {$q$} (D);
\end{tikzpicture}
```

See



Design

Every ordinary mapping shares one arrow style. Label position changes with path direction, because placement serves readability rather than uniform syntax.

10.9 Debugging Workshop

10.9.1 Forgetting Braces Around a Named Tip

Preferred modern syntax:

```
\draw[-{Stealth}] (A) -- (B);
```

10.9.2 Putting the Label Outside the Path

Incorrect:

```
\draw[-{Stealth}] (A) -- (B);  
\node {$f$};
```

Correct:

```
\draw[-{Stealth}] (A) -- node[above] {$f$} (B);
```

10.9.3 Using sloped Automatically

Judge the result. Keep a label horizontal when rotating it makes the text harder to read.

10.9.4 Using Too Many Tip Designs

If arrowheads do not represent distinct mathematical meanings, replace them with one shared convention.

10.10 Exercises

1. Draw a Stealth arrow and a Latex arrow for comparison.
2. Put Stealth tips at both ends of a segment.
3. Place labels at positions .25, .5, and .75 on three paths.
4. Put two labels on opposite sides by using `auto` and `swap`.
5. Add a sloped label to a curved arrow.
6. Define a shared style for every mapping in a diagram.
7. Draw two opposite curved arrows with separate readable labels.
8. Add a fifth labeled arrow to the worked project.

10.11 Solutions

10.11.1 Exercises 1–4

```
\draw[-{Stealth}] (0,0) -- (4,0);  
\draw[-{Latex}] (0,-1) -- (4,-1);
```

```
\draw[{Stealth}--{Stealth}] (0,0) -- (4,0);
```

```
\draw[->] (A) -- node[pos=.25,above] {$f$} (B);  
\draw[->] (A) -- node[midway,above] {$g$} (B);  
\draw[->] (A) -- node[pos=.75,above] {$h$} (B);
```

```
\draw[->] (A) -- node[auto] {$f$} (B);  
\draw[->] (A) -- node[auto,swap] {$g$} (B);
```

10.11.2 Exercises 5–7

```
\draw[-{Stealth}] (A) to[bend left=25]  
node[midway,sloped,above] {$f$} (B);
```

```
\tikzset{map/.style={-{Stealth},thick}}
```

```
\draw[map] (A) to[bend left=25]  
  node[above] {$f$} (B);  
\draw[map] (B) to[bend left=25]  
  node[below] {$g$} (A);
```

For Exercise 8, preserve the four node positions and the shared `map` style; add only one path and its label.

10.12 ChatGPT Workshop

Ask ChatGPT to preserve the mathematical relationships while checking arrow and label choices.

1. Replace every old arrow tip in this diagram with one consistent **arrows.meta** convention.
2. Move these labels with **pos**, **auto**, or **swap** without changing any node coordinates.
3. Explain whether **sloped** improves each curved label in this figure.
4. Find every label that was written outside its path command.
5. Identify arrow-style differences that do not communicate mathematical meaning and simplify them.

10.13 Final Checklist

Before continuing to Chapter 11, check that you can

1. load and use modern arrow tips;
2. place tips at the start, end, or both ends;
3. position labels with verbal and numerical options;
4. place labels on opposite sides of a path;
5. decide whether a curved label should be sloped;
6. define one shared arrow convention; and
7. keep every path label inside its path command.